



Genome-Wide Identification and Analysis of Chromosomally Integrated Putative Prophages Associated with Clinical *Klebsiella pneumoniae* Strains

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Abstract

Klebsiella pneumoniae, an opportunistic pathogen found in the environment and human mucosal surfaces, is a leading cause of nosocomial infections. *K. pneumoniae* is now considered a global threat owing to the emergence of multidrug-resistant strains making its infections untreatable. In this study, 254 strains of *K. pneumoniae* were screened for the presence of prophages using the PHASTER tool. Very few strains lacked prophages (3.1%), while the remaining harboured both intact (811) and defective prophages (709). A subset of 42 unique strains of *K. pneumoniae* was chosen for further analysis. Our analysis revealed the presence of 110 complete prophages which were further classified as belonging to *Myoviridae* (67.3%), *Siphoviridae* (28.2%) and *Podoviridae* family (4.5%). An alignment of the 110 complete, prophage genome sequences clustered the prophages into 16 groups and 3 singletons. While none of the prophages encoded for virulence factors, 2 (1.8%) prophages were seen to encode for the antibiotic resistance-related genes. The CRISPR-Cas system was prevalent in 10 (23.8%) out of the 42 strains. Further analysis of the CRISPR spacers revealed 11.42% of the total spacers integrated in *K. pneumoniae* chromosome to match prophage protein sequences.

Introduction

Klebsiella pneumoniae is one of the most significant opportunistic pathogens responsible for nosocomial infections among critically ill and immunocompromised individuals [1]. *K. pneumoniae* is the causative agent of several life-threatening conditions, like urinary tract infection, pneumonia, intra-abdominal infection, bloodstream infection, meningitis and pyogenic liver abscess [2]. The increasing clinical incidence of multidrug-resistant (MDR) *K. pneumoniae* strains is a growing worldwide problem as there is a limited choice of an effective antibiotic for treatment. *K. pneumoniae* has been shown to harbour prophages within its genome, the presence of which has been correlated to its multidrug resistance [3] and contributing to its genome plasticity [4] in this pathogen. Furthermore, *K. pneumoniae* is an industrially important organism owing to its nitrogen

fixation, bioremediation and bioconversion capabilities. The presence of prophages in industrial strains could be detrimental to the fermentation industry [4].

Phages being the most abundant organisms on earth are the major contributors in shaping the bacterial communities in most ecosystems [5, 6]. While lytic bacteriophages parasitize and lyse the host, prophages are integrated into viral forms that can reside in host chromosomes or plasmids until the conditions favour their reactivation [6]. Integrated prophages within the bacterial genome have been linked to host fitness, increased infection, antibiotic resistance, pathogenicity and virulence in several bacterial species [5, 7–9]. The presence of lysogenic phage may also provide a selective advantage to the host bacterium against secondary phage attack [10]. Several recent studies have been focused on the characterization and comparative analysis of the prophage within pathogen genomes, such as *Acinetobacter baumannii* [11]; *Burkholderia cenocepacia* [12]; *Helicobacter pylori* [13]; *Mycobacterium* [14]; *Streptococcus pneumoniae* [7] and *Streptococcus mutans* [15].

Klebsiella species are polylysogenic and considered interesting models to study the role of prophages [4]. Further, it is known that in the clinical setting many antibiotics stimulate prophage induction and facilitate phage infection [16].

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Thus, in this study, we aimed to look into the prevalence of chromosomally integrated prophages within clinical *K. pneumoniae* strains and characterize them based on their diversity and the presence of major phage genomic elements. The possible associations of the antibiotic resistance and virulence genes encoded by the integrated phage if any were also looked into.

Materials and Methods

Sequence Retrieval and Prophage Prediction

A dataset of 254 complete genomes of clinical origin available for *K. pneumoniae* as of February 2019 was retrieved from the GenBank database (<https://www.ncbi.nlm.nih.gov/>) for prophage analysis. Plasmids were not included in this study. All the strains were screened for prophage-like elements using the webserver PHASTER, a program that enables the rapid identification and annotation of prophages [17]. The PHASTER tool employs a scoring system to assesses the completeness of the prophage and the output is given as intact (score > 90), Questionable (score 70–90) or incomplete (score < 70). To further eliminate the false positives, the intact phages were manually curated for their completeness by looking for the presence of all intact prophage-associated genes, such as integrase, endolysin and terminase genes, and ‘att’ sites (attL/attR) [18]. Putative prophages lacking any of these genes were considered degenerate and eliminated. Similarly, questionable and incomplete prophages were also manually curated and if defective were eliminated. The chromosomal sequences harbouring only intact prophages were used in generating a dendrogram that grouped the sequences into 42 clusters. Strains within each subcluster having the maximum number of putative intact prophages and a higher genomic coverage were further selected for prophage analysis. The associated MLST type for each strain was identified using the PATRIC database (<https://www.patricbrc.org/>). The capsular serotypes were computationally predicted using the K-PAM (https://www.iith.ac.in/K-PAM/serotype_pdb.php) a *Klebsiella* species serotype predictor and surface antigens modeller tool [19].

Prophage Classification

The identified complete prophages were classified into their respective phage families based on the protocol described earlier [11]. The BLASTp tool (version 2.8.0+) was used to search for available phage sequences in the NCBI database using default parameters. The phage elements considered indicative of a family, such as the major capsid proteins, tail proteins and large terminase subunit were further classified into their respective prophage families with BLAST

search parameters set to E -value < 1×10^{-7} , cut off identity value > 35% and a coverage > 50%.

Comparative Genomics of Prophage Sequences

All identified complete prophage sequences were aligned using the MAFFT Version 7.0 program (<https://mafft.cbrc.jp/alignment/server/>), using the strategy ‘auto’. The phylogenetic tree of the genomes was subsequently constructed using the phylogeny program (<https://mafft.cbrc.jp/alignment/server/phylogeny.html>) using the neighbour-joining method using bootstrap values of 1000 replicates [20]. The generated phylogenetic tree was visualized using the iTOL [21] program.

Prophage-Associated Virulence Factors and Antibiotic Resistance Genes

A Blastn search (100% coverage, E -value < 0.0001) against the Virulence factor database (<http://www.mgc.ac.cn/VFs/blast/blast.html>) core dataset A and B [22] was used to identify putative virulence factors harboured by the integrated prophages within *Klebsiella* strains. Similarly, the prophage genomic sequences were screened for antibiotic resistance genes using the resistance gene identifier (RGI) option of the comprehensive antibiotic resistance database (<https://card.mcmaster.ca/analyze/rgi>) using default values [23].

Identification of CRISPR-Cas System in *K. pneumoniae*

Klebsiella pneumoniae genome confirming for the presence of complete putative prophages were also studied for the presence of CRISPR-Cas systems using the tool CRISPRCas-Finder <https://crisprcas.i2bc.paris-saclay.fr/CrisprCasFinder/Index> [24]. Putative origins of spacers were inferred using the nucleotide BLAST tool of NCBI with search parameters set to default values.

Statistical Analysis

The correlation between the integrated putative intact prophage genome sequence sizes to the corresponding *K. pneumoniae* genomic size was tested using the Spearman correlation coefficient test. A P value < 0.05 was considered significant. The statistical inferences and graphs were generated using GraphPad Prism version 6.0 (San Diego California USA, www.graphpad.com).

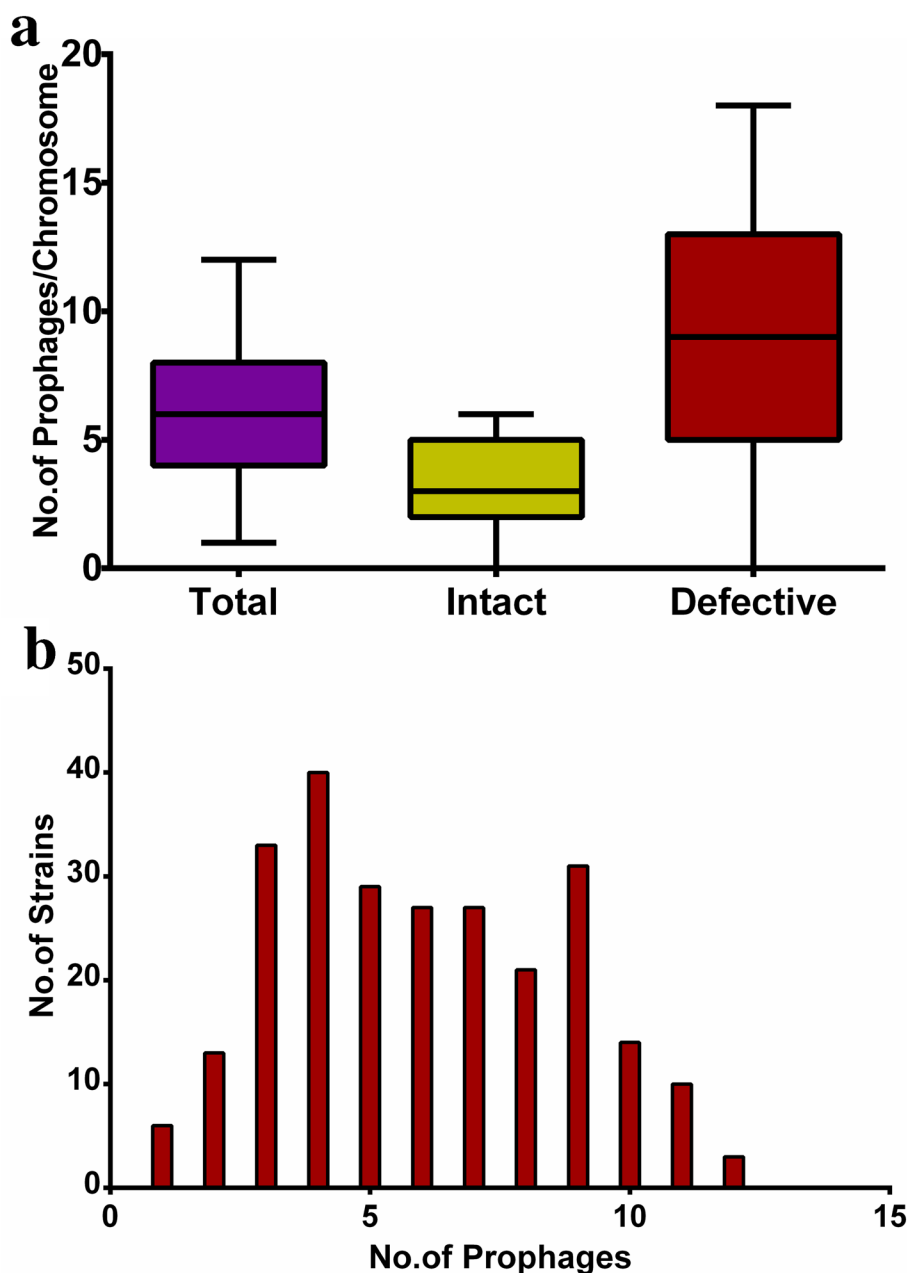
Results

Identification and Classification of Prophages

In this study, of the 254 *K. pneumoniae* strains analysed for prophages using the PHASTER program (accessed February, 2019), 8 (3.1%) strains lacked putative intact prophages within their genomes. A total of 1520 prophages were detected in the remaining strains, of which 811 were classified as intact and 709 as defective. The total number of prophages per genome ranged from 1 to 12, with an average being 5.98 (Fig. 1a). While a majority of the

strains were observed to harbour either four ($n=40$), three ($n=33$) and nine ($n=31$) prophages, few of the strains ($n=27$) were also seen to have more than ten prophages (Fig. 1b) integrated into the chromosomes. Three Chinese *K. pneumoniae* isolates SCKP020079, F44 and WCHKP040035 carried up to 12 prophage-like regions in the chromosome with the prophages making up 7.55%, 6.33% and 7.56% of the total genomic content, respectively (Supplementary dataset S1). The genome sizes of the *K. pneumoniae* strains ranged from 5.1 to 6.4 Mb. A Spearman correlation test to assess whether larger genomic size favoured the integration of more phages indicated a positive correlation (Spearman's $\rho=0.27$,

Fig. 1 **a** Box plot depicting the total, intact and defective prophages in the chromosomes of *K. pneumoniae*. The horizontal line at the centre of the plot represents the median. The bottom and top of the plot represent the first and third quartiles. The external edges of the plot represent the minimum and maximum number of prophages per genome. **b** Bar graph depicting the distribution of total prophages (intact and defective) in strains of *K. pneumoniae*



P value <0.0001). The 254 strains could be classified into 54 different MLST types and 60 capsular serotypes. Most strains belonged to the MLST types ST258 (16.1%), ST11 (15.74%), 147 (8.66%), ST340 (5.12%) and ST14 (4.72%). Further, it was seen that the high-risk epidemic clones from China harbouring between 7 and 12 prophages were of the MLST type ST11 and the capsular serotypes K47 and K64. Similarly, the MLST ST258 type consisted of two capsular serotypes KL107_variant (mostly of USA origin) and KL106 (strains from the mixed geographical origin) and carried 6–10 prophages. The capsular serotypes K64 and K15 belonging to the MLST type ST147 and ST340 (strains from the mixed geographical origin) harboured around 1–5 and 6–7 prophages, respectively. The K2 capsular serotype was the most dominant among ST14 (strains from the mixed geographical origin) and carried 3–6 prophages (Supplementary Table 1).

A subset of 42 genomes that was chosen for further analysis yielded a total of 110 complete prophage-like elements, details of which are given in Supplementary Table 1. The prophages within these genomes ranged from 1 to 6 occupying a genomic content ranging between 0.7 and 5.4% of the host bacterium. The average complete prophage genomic size and G+C% was around ~47,000 bp and 52.1%, respectively (Supplementary Table 1). A genomic view of a

representative prophage strain indicating the features are presented in Fig. 2. Genomic analysis of upstream and downstream regions of the prophage attachment sites (attL and attR) revealed 21 (19%) prophages to have tRNA integrated before and after the intact prophage and tRNA–arg to be the most common. The other integration ‘hotspots’ commonly identified were the transcriptional regulators (25 prophages, 22.7%) and the ABC transporters (11 prophages, 10%) (Supplementary Dataset S1).

All of the complete prophages identified could be assigned to a family with a majority 74 (67.3%) assigned to the *Myoviridae* family, 31 (28.2%) to *Siphoviridae* and 5 (4.5%) to the *Podoviridae* family. The *Myoviridae*, *Siphoviridae* and *Podoviridae* prophages had an average genomic sizes of 47.08 kb, 47.58 kb and 47.48 kb, respectively. It was also seen that several (~40%) of the host *K. pneumoniae* strains carried mixed phage species within their genome (Supplementary Table 1).

Phylogenetic Relationships Between Prophages

To determine the diversity of the prophages identified, a phylogenetic tree was generated based on complete prophage sequences. Results showed the prophages to be diverse being distributed within 16 clusters and three singletons (Fig. 3).

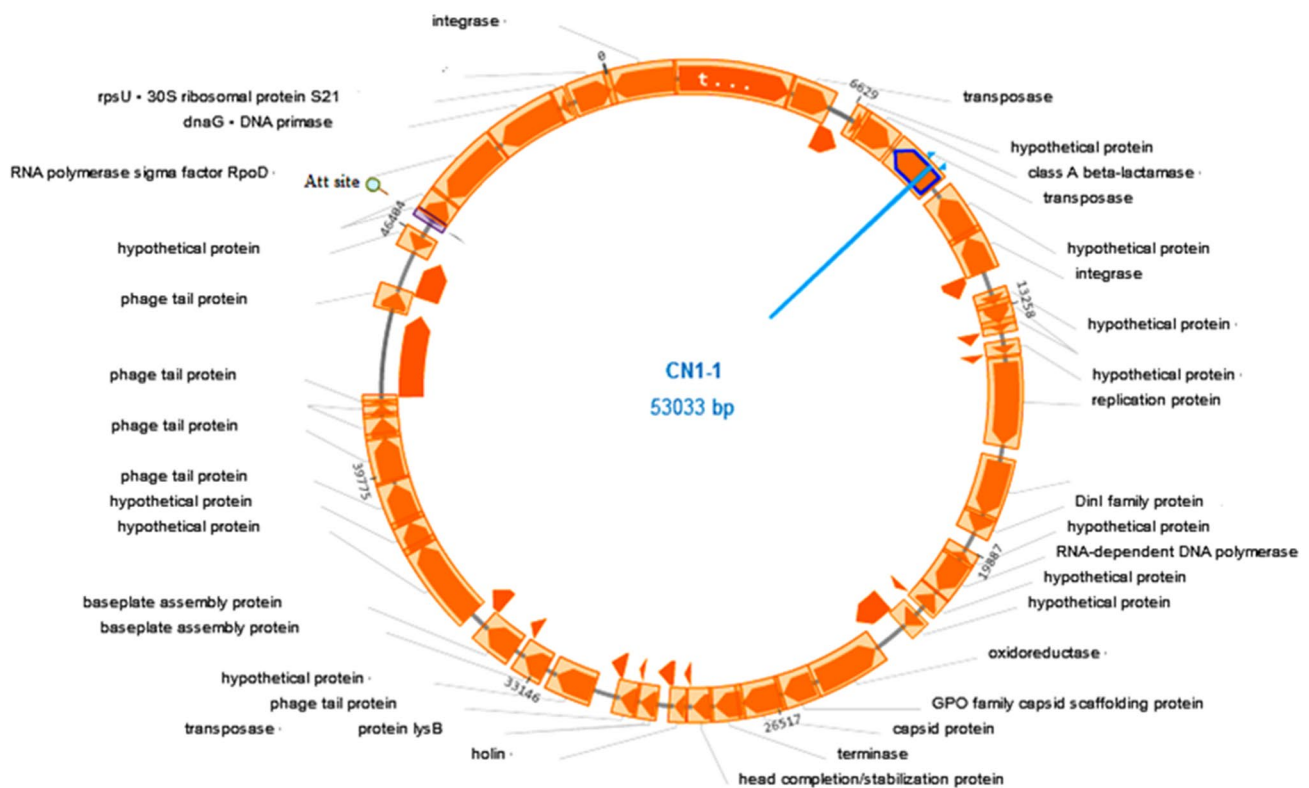


Fig. 2 Genomic map of a putative prophage within the genome of the *Klebsiella pneumoniae* strain CN1-1 (Acc. No CP015382.1). The map was generated using the Genome Compiler tool

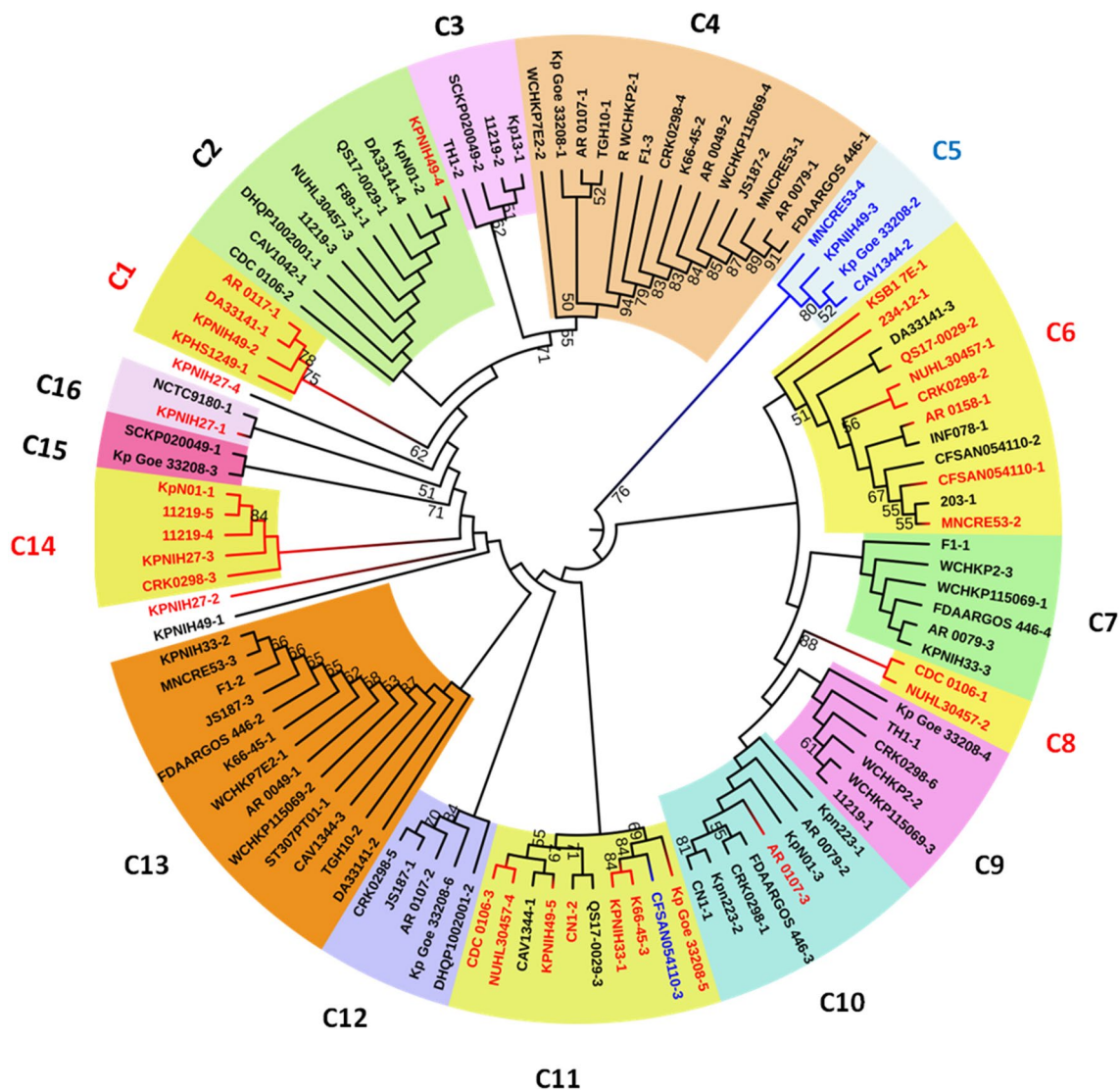


Fig. 3 Phylogenetic tree of *Klebsiella pneumoniae*-associated prophages. The neighbour joining genomic tree of putative intact prophages was generated using MAFFT program with 1000 Boot-

strap replicates. Bootstrap values >50% are indicated. The *Myoviridae*, *Siphoviridae* and *Podoviridae* prophages are indicated as black, red and blue, respectively (Color figure online)

The 74 *Myoviridae* prophages were seen to group exclusively into seven clusters (C3, C4, C7, C9, C12, C13, C15) and one singleton. Similarly, the *Siphoviridae* were grouped exclusively into three clusters (C1, C8, C14) and two singletons. Among *Podoviridae* four out of the five phages clustered together in a single group (C5). Clusters C1, C6, C10, C11 and C16 had mixed phages (Fig. 3).

Presence of Virulence Factors and Antibiotic Resistance Genes Within Prophages

Prophage regions analysed for the presence of virulence factors revealed the absence of any of the virulence-associated genes. However, it was observed that around 10 (9%) of

the prophages had closely integrated T6SS-associated genes along the upstream and downstream regions of the prophage attachment sites (Fig. 4).

Only two prophages (1.8%) were found to encode antibiotic resistance genes. The putative intact prophage from the ST392 strain CN1 encoded KPC beta-lactamase and the putative intact prophage from the ST512 strain CFSAN054110 encoded the antibiotic genes sulfonamide-resistant dihydropteroate synthase *Sul1*, ANT (3')-Ia family aminoglycoside nucleotidyltransferase *AadA2* and putative quaternary ammonium compound-resistance protein *qacE* (Fig. S1). In the prophage region of the strain CN1 carrying the beta-lactamase gene, a transposase gene was located 1kbp upstream and 250 bp downstream. Similarly,

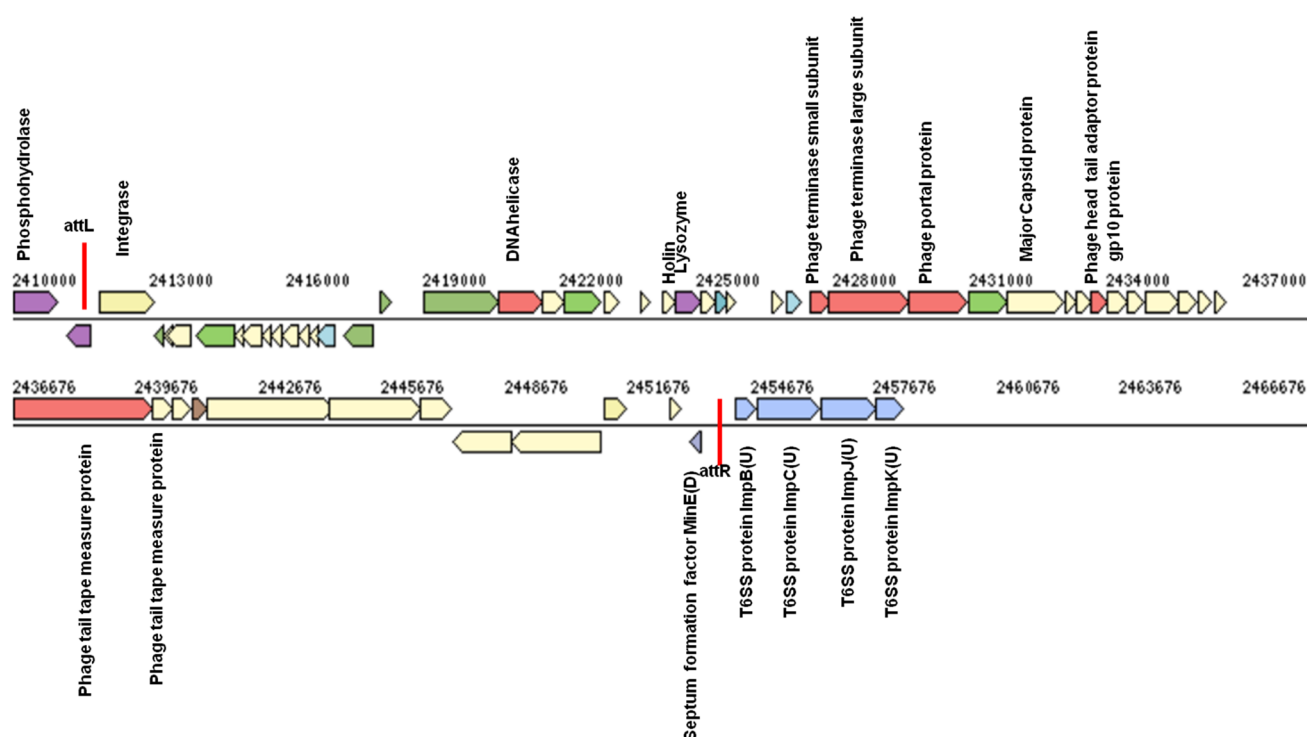


Fig. 4 Genomic context of the intact prophage of the *K. pneumoniae* strain 11,219 in relation to its Type 6 Secretion System component

IS6 family transposase was found ~2kbp upstream of the antibiotic resistance genes encoded by the prophage region of the strain CFSAN054110. The IS5-like element was found ~16 kb downstream of this region.

CRISPR-Cas and Spacer Analysis in *Klebsiella* Genomes

In this study, only nine out of the 42 (21.4%) strains were seen to harbour the complete CRISPR-Cas system of the type 1E. Of these, three strains possessed two CRISPR

arrays, while one strain was seen to carry an orphan CRISPR system lacking the presence of associated Cas genes (Table 1). The prophages and CRISPR-Cas system harboured on the chromosome was separated by a proximal distance of ~50 kb nucleotides. Analysis of spacer content revealed a total of 324 spacer sequences of which 37 (11.42%) matched to prophage sequences carried by other strains. However, none of the spacers was found to be self-targeting. Further analysis of the spacers revealed that they were homologous to phage proteins, like integrase, capsid protein, tail protein, terminase, head morphogenesis protein,

Table 1 CRISPR and spacers identified in *K. pneumoniae* strains analysed in this study

Strain	No. of prophages	CRISPR array	Type	Number of spacers	Spacers matching prophages from other strains
AR_0117	1	2	1E	6, 8	2,0
TGH10	2	1	1E	24	2
AR_0158	1	1	1E	52	5
CDC 0106	3	2	1E	9, 12	1,0
Kpn223	2	1	1E	43	5
KPNIH27	4	1	1E	39	6
CN1	2	1	1E	39	6
CAV1344	3	1	1E	42	3
11219	5	1	Orphan	3,16	1
KPHS1249	1	2	1E	5,26	2,4

helicases, methyltransferases and several hypothetical proteins encoded by the *Klebsiella* prophages (Fig. 5).

Discussion

Prophages are horizontally transferred genetic elements that significantly contribute to the genome variability, evolution and virulence of bacterial hosts [25]. Our initial study with 254 *Klebsiella* strains showed that very few strains lacked prophages (3.1%), while the remaining harboured both intact and defective prophages. The presence of such defective prophages has been reported earlier [26] and thought to have emerged due to deletion or disruption of genes, and helping the bacterium cope up with adverse environments [27]. However, in this study, we carried out a systematic characterization of complete putative prophages for a subset of 42 *K. pneumoniae* strains, their genomic organization and phylogenetic analysis. The 42 strains analysed were seen to have prophages integrated into their chromosomes with several showing the presence of mixed prophage types. Such observations have been reported earlier for several pathogens like *E. coli* [28], *Streptococcus pneumoniae* [7], *Bacillus thuringiensis* [29], *Acinetobacter baumannii* [11] and *Clostridioides difficile* [30]. The presence of multiple prophage-like elements indicates that polylysogeny could be a common feature among several bacterial pathogens, including *K. pneumoniae*. Previous reports have attributed polylysogeny

to increased bacterial host fitness [31, 32], acquisition of multiple pathogenic traits and to an increase in the pathogenicity of the bacterium, which in turn could provide an ecological and evolutionary benefit to the phage [32]. The % G+C content of the prophage genome in this study was consistently lower than that of the host genome (Supplementary Data S1), indicating the probability of horizontal gene transfer events [33] within the host. Our initial observations revealed that a substantial percentage (61.94%) of prophages within the *K. pneumoniae* genomes to be defective. Such defective phages within bacterial genomes have been reported earlier, their presence being attributed to selective pressure exerted by the host bacterium on the newly acquired phages resulting in inactivation mutations, leading to deletions or disruptions of genes essential to produce infectious particles [11, 26]. Such mutations could also offer multiple benefits to the host for coping with adverse environmental conditions [27]. Further, in bacteria, lysogenization has been correlated to its genomic size and minimal doubling time [34]. Bacteria with a minimal doubling time <2.5 are considered fast growers and shown to be more lysogenized than slow growers [34]. Doubling time for *K. pneumoniae* has been reported to be ~0.4 [35]. Thus, as a fast grower, a higher percentage of phages observed for *K. pneumoniae* in this study is as expected [11] assuming significance. Further, studies suggest that the presence of lysogens make these otherwise fast-growing bacteria to grow weakly under hostile environmental conditions, such as those encountered in

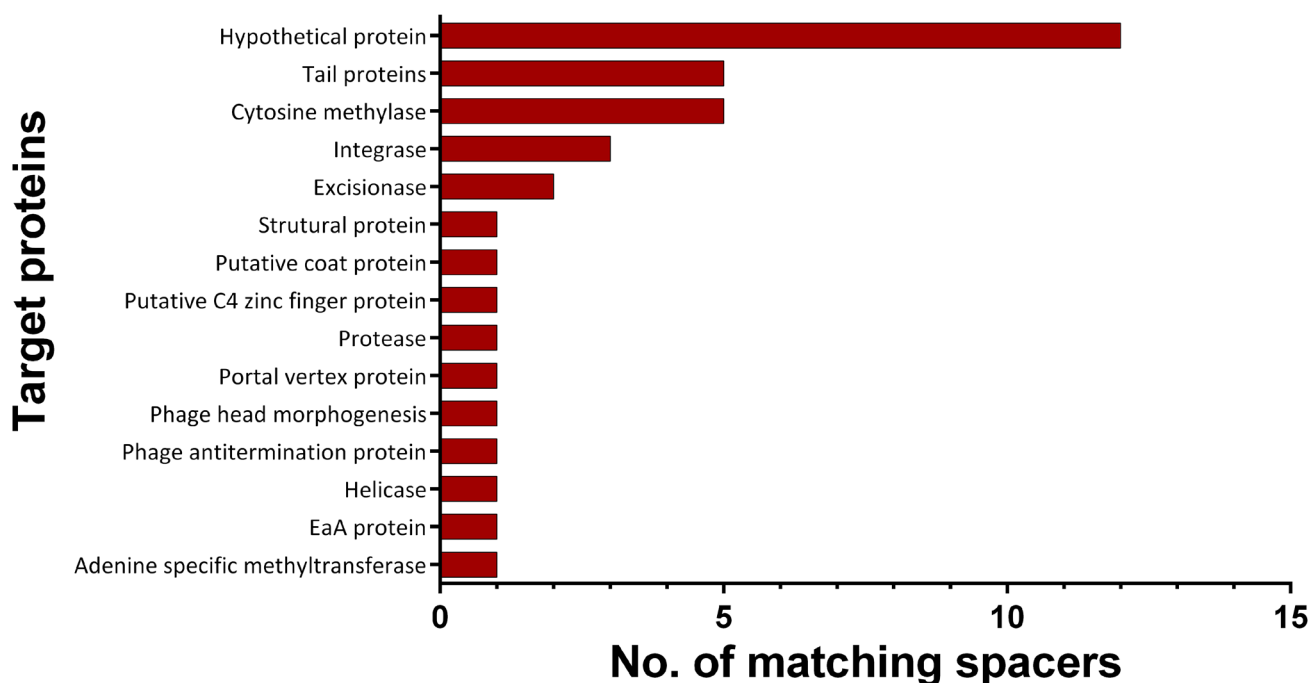


Fig. 5 CRISPR spacers matching the target proteins in the putative complete prophages of *K. pneumoniae*

hospitals, to preserve their genome while waiting for ideal conditions for their lytic propagation [36].

Reports suggest that prophages occupy 10–20% of a bacterium's genome [37]. *E. coli* O157:H7 str. Sakai harbours 18 phages constituting 16% of its total genomic size [26]. *K. pneumoniae* in this study was seen to harbour phages ranging from 1 to 6 occupying a genomic content of 0.1–8% which is in agreement with an earlier report on *K. pneumoniae* [4]. A positive correlation between the numbers of prophages to genomic size in *K. pneumoniae* strains studied is in accordance with an earlier report on this bacterium [16].

In *K. pneumoniae* genomes, the phages belonging to the *Myoviridae* family were highly represented (67.3%). Similarly, the *Myoviridae* phages accounted for most of the prophages (69.2%) in the *Desulfovibrio* genomes [38]. However, *Siphoviridae* formed the largest group (67%) in the genome of *Acinetobacter baumannii* [11]. The genome sizes of the prophages analysed in this study fell within the range specified for *Myoviridae* (30.3–79.0 kb), *Podoviridae* (46.4–68.0 kb) and *Siphoviridae* (37.8–88.3 kb) [39]. The 'att sites' of varying lengths were found in the identified prophages. These regions are the 'hot spots' for site-specific recombination brought about by the integrases. The huge variation in the length of 'att sites' and the extent of duplication depends on the phage-type and integration site within the host genome [40]. Our results indicated tRNA and transporters, such as the antimicrobial peptide ABC transporter permease SapC, to be major prophage integration hotspots which is consistent with earlier findings [41].

All identified prophages were also found to encode for the protein terminase, which function as site-specific endonucleases, involved in cleaving phage DNA before packaging during the maturation step [42].

The phylogenetic tree based on the whole genome analysis revealed that phages from the same strain clustered separately, indicating the prophage evolution to be independent of the host. Furthermore, phage genomes are often shown to have a mosaic nature owing to the horizontal transfer and recombination events taking place during integration and subsequent excision from the host genome [43]. Phage integrases are essential for the integration and excision of the prophage genome into the bacterial host. They serve as diagnostic markers for prophages in the bacterial genome [14]. Endolysins and holins together constitute the lysis cassette of the bacteriophages and are essential for programmed cell lysis upon completion of bacteriophage life cycle [44]. The presence of the endolysin gene indicates the infective ability of the phages upon induction. Cloning and expression studies of these lytic genes would have major implications in controlling this multidrug-resistant pathogen. Only 25 prophages out of

the total 110 prophages encoded the holin gene. Certain endolysins have been reported to function in holin-independent manner [44].

Recent studies have implicated that bacteriophages residing within bacterial genomes are intimately associated with virulence and the evolution of several bacterial pathogens [45]. In this study, few of the *Klebsiella*-associated prophages were observed to encode for antibiotic resistance genes. This assumes significance as prophages play an important role in mediating the transfer of antibiotic resistance genes to bacteria, thereby increasing their environmental tolerance [29]. An earlier study involving the human gut microbiome indicated that phages could act as vehicles of ARG transfer [4]. Thus, the presence of transposons flanking the antibiotic resistance genes observed in two of the strains probably indicates a possible role in their horizontal transfer. However, it remains unclear whether these resistance genes are transferred to the current location by the resident prophage or by the action of the transposable elements [4]. Although a few studies have indicated that prophage elements function as reservoirs for virulence genes [11], no such observations were made for *Klebsiella* prophage elements in this study. However, few of the *Klebsiella* prophages were seen to harbour T6SS genes both upstream and downstream of prophage attachment sites. The exact association of these genes to prophage is unknown.

Also, CRISPR/Cas system analysed in this study showed it to be of Type IE, located within the putative prophage(s) of few strains. Analysis of the CRISPR spacers showed it to be homologous to several prophage sequences. Similarly, the analysis of spacer sequences of *Cronobacter sakazakii* showed that it matched the prophages from the host strain as well as from other strains [46]. Considering that the function of the CRISPR/Cas system is to destroy invading DNA, it could probably be hypothesized that such prophages could protect the bacterium from further invasion of similar phages [47]. An analysis of the CRISPR spacers identified indicated that the spacers were not self-targeting which probably led to the tolerance to integrated prophages thereby contributing to strain diversification. A recent study also shows the limited impact of CRISPR-Cas system on the prophages in *K. pneumoniae* [4]. This warrants further studies on the interaction of CRISPR-Cas systems with prophages and their contribution towards bacterial fitness. In summary, our results highlight that the clinical *K. pneumoniae* harbour a relatively high number of prophages. A recent study shows that prophages significantly contribute to genome plasticity in *K. pneumoniae* [4]. Our results are in agreement with this study. A relatively high number of prophages might play a significant role in bacterial evolution, adaptation and physiology. Further, prophage induction experiments need to be carried out to clearly understand host–phage interactions.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00284-021-02472-2>.

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Author Contributions MS, PB and GSK designed this study. PB carried out the analysis work and drafted the manuscript. MS was involved in data curation and supervision, reviewing and editing of the manuscript. GSK was involved in funding acquisition and supervision. All the authors read and approved the final manuscript.

Declarations

Conflict of interest All the authors declare that they have no conflict of interest.

Ethical Approval This article does not contain any research involving humans or animals.

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